Paleomagnetism of the Late Proterozoic sills in the Tsezotene Formation, Mackenzie Mountains, Northwest Territories, Canada[1]

J. K. PARK

Geomagnetic Laboratory, Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa, Ont., Canada K1A 0Y3

Received January 6, 1981

Revision accepted May 29, 1981

A paleomagnetic study was conducted on a regionally extensive diabase sill intruding the upper part of the Helikian (?) Tsezotene Formation located in the Mackenzie Mountains Supergroup of northwestern Canada. Twelve sites, one from a lower sill at one locality, were sampled over a distance of about 200 km. Two general magnetizations presumably carried by maghemite (A_R: 38 specimens; 272°, 00°; pole 139°E, 01°N; $K = 34$, $A_{95} = 4°$) and hematite (B: 18 specimens; 269°, −01°; pole 141°E, 01°S; $K = 27$, $A_{95} = 7°$) were resolved by using a two-stage treatment method (alternating field treatment of 20–45 mT followed by thermal demagnetization to about 650°C) in combination with vector subtraction and vector diagrams. B has normal (B_N: 6 specimens) and reverse (B_R: 17 specimens) components. Another magnetization (R_M: 30 specimens; 086°, +55°; $k = 3$) carried by magnetite appears to be a resultant of normal and reverse components, though it may include a secondary Cretaceous component. It is likely that the maghemite and hematite are *deuteric* alteration products of the magnetite and that they retain the primary direction probably residing in the unresolved magnetite components of R_M.

Une étude paléomagnétique a été effectuée dans une région comprenant de nombreux filons-couches de diabase pénétrant la partie supérieure de la formation Tsezotene de l'Hélikien (?) située dans le supergroupe des montagnes Mackenzie du nord-ouest du Canada. Douze sites ont été échantillonnés sur une distance d'environ 200 km dont un situé plus bas du filon-couche provient d'un à une localité. Deux aimantations générales relevant probablement de la maghémite (A_I: 38 spécimens; 272°, 00°; pôle 139°E, 01°N; $K = 34$, $A_{95} = 4°$) et de l'hématite (B: 18 spécimens; 269°, −01°; pôle 141°E, 01°S; $K = 27$, $A_{95} = 7°$) ont été reconnues en utilisant une méthode de traitement en deux étapes (champ alternatif de 20 à 45 mT suivi de la désaimantation thermique jusqu'à environ 650°C) en conjonction avec la soustraction des vecteurs et les diagrammes de vecteurs. B possède une composante normale (B_N: 6 spécimens) et une composante inverse (B_I: 17 spécimens). Une aimantation additionnelle (R_M: 30 spécimens; 086°, +55°, $k = 3$) portée par la magnétite semble être la résultante de composantes normale et inverse bien qu'elle puisse inclure une composante secondaire d'origine Crétacé. La maghémite et l'hématite apparaissent comme des produits de l'altération *deutérique* de la magnétite et ces minéraux probablement conservent la direction primaire résidant probablement dans les composantes non-résoutes (R_M) de la magnétite.

[Traduit par le journal]

Can. J. Earth Sci., **18**, 1572–1580 (1981)

Introduction

The stratigraphic succession within the Mackenzie Mountains Supergroup of the Canadian Cordillera is given in Table 1. An unconformity above the Little Dal Group presumably divides this Precambrian succession into Helikian and Hadrynian age sequences. All of these units to at least as high as the "grainstone subunit" of the Little Dal Group are cut by diabase intrusions, which are thought to be lithologically equivalent to the diabase sills sampled for the present study within the Tsezotene Formation (Aitken and Cook 1974). The porphyritic basaltic Little Dal lavas occurring at the top of the Little Dal Group are probably unrelated to the sills (Aitken 1980; Morris and Park 1981). The sills are found only in the Tsezotene Formation, predominantly in the upper 150 m. In some areas three or more sills are present, but one major sill ranging from about 18 to 60 m thick appears to be of regional extent (Aitken and Cook 1974). The sills along with the other Proterozoic units were folded in Paleocene or pre-Late Cretaceous times and have suffered extensive low-grade alteration.

Regional low-grade alteration has affected all the potassium minerals in the sills. This effect, together with the low bulk potassium content of these rocks, rendered early attempts at dating them unsuccessful (D. K. Norris *in* Aitken and Cook 1974). Recent Rb–Sr determinations on two of the sills have yielded minimum ages of 766 ± 24 and 769 ± 27 Ma ($\lambda = 1.42 \times 10^{-11}$ year^{-1}; R. L. Armstrong, G. H. Eisbacher, and P. D. Evans, in preparation; J. D. Aitken, personal communication, 1981).

Methods

The regional sill in the Tsezotene Formation was sampled at 11 sites from several localities spread over a distance of about 200 km (Fig. 1). Five oriented hand

[1]Contribution from the Earth Physics Branch No. 945.

0008-4077/81/101572-09$01.00/0

©1981 National Research Council of Canada/Conseil national de recherches du Canada

TABLE 1. Table of formations

Mackenzie Mountains Supergroup
Coppercap Formation
Redstone River Formation
Little Dal lavas
Little Dal Group
Katherine Group
Tsezotene Formation
H1 unit

samples were generally obtained at each of these sites. At an additional site (1) from a 17.5 m thick lower sill, approximately 690 m below the major sill, oriented drill cores were obtained. Hand samples were cored in the laboratory and all cores were sliced into two or three specimens.

Specimens were treated by alternating fields (AF's) and by heat within spaces having low residual fields of usually <2 nT (Roy *et al.* 1969). Most specimens were treated by a two-stage method involving AF treatment followed by thermal. AF work was carried out on apparatus described by Roy *et al.* (1973) and thermal work on a Schonstedt TSD-1 furnace. Specimens were measured on Schonstedt DSM-1 spinner magnetometers.

The complex magnetization was analyzed by vector subtraction (VS) (Irving *et al.* 1974; Park and Roy 1979) and by vector diagrams (VD's) (Zijderveld 1967). VS can usually only be profitably used when it has been determined by VD's or other means that essentially only

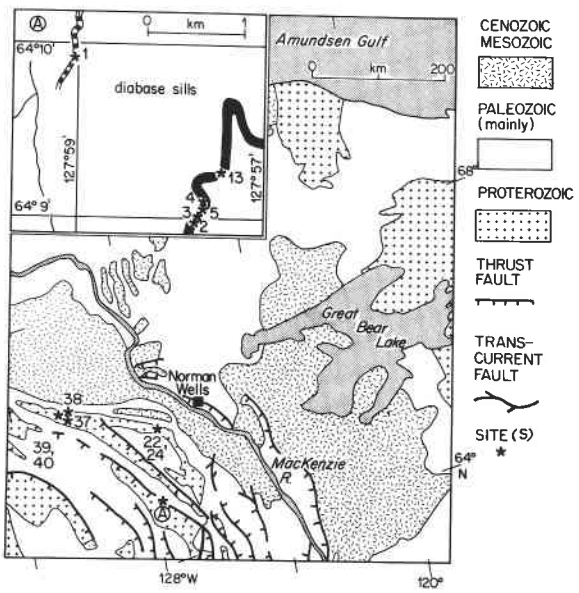


FIG. 1. Site location map. Site coordinates are given in Table 2.

one component of the magnetization is being eliminated within a given treatment interval. There is at least one case, however, where VS can resolve *two* magnetizations that are being concurrently removed within a given treatment interval, namely, the instance where the two magnetizations are antiparallel to within about 20°. They may even have nearly complete overlap of unblocking temperature (T_{UB}) or resistive coercive force (rcf) spectra. This special case, found in the present study, commonly results from reversals of the Earth's magnetic field during the acquisition of magnetization. The technique for analyzing this case has been fully explained in Park and Roy (1979). It relies on small relative changes between the rates of decay of the components. Because the resultant of nearly antiparallel components having similar magnitudes is generally small compared with its constituents, these changes can produce relatively large changes in the direction and intensity of the resultant. As the resultant magnetization approaches in direction one or the other of its component directions, its intensity and homogeneity (related to the internal precision of the measurement) increase. The best approximation to the true direction irrespective of sign is found by subtraction between vectors that have relative maximum or minimum values of intensity or direction on intensity decay curves, stereoplots, or vector diagrams. Any one determination will not yield a true direction unless only one of the components changes, but an average of many VS directions should approach the true direction. The use of VD's will not directly resolve this case except to indicate which vectors are to be subtracted.

Results and analysis

During the course of the analysis about 50% of the collection or some 50 specimens were eliminated from the study; their magnetizations were either composed of unresolvable components or they had very high intensities up to 300 A/m, which could have been induced by lightning.

Natural remanent magnetizations (NRM's) of the remaining specimens had average intensities of about 850 mA/m and scattered directions of mainly positive inclination (Fig. 2). Initial treatment of these NRM's by AF experiments eliminated magnetizations having remanent coercive forces ranging up to 50 mT (Figs. 3, 4) but quite scattered directions (Fig. 5). Thermal treatment of 10 specimens eliminated a northerly directed magnetization in four of them with T_{UB}'s between 200 and 550°C (Fig. 5). Typical thermal intensity decay curves are shown in Fig. 6. Because both AF and thermal directions are somewhat scattered (combined: $N = 52$, $k = 3$), though yielding a significant combined vector resultant (mean 292°, +56°; $R = 32.5$, $\alpha_{95} =$

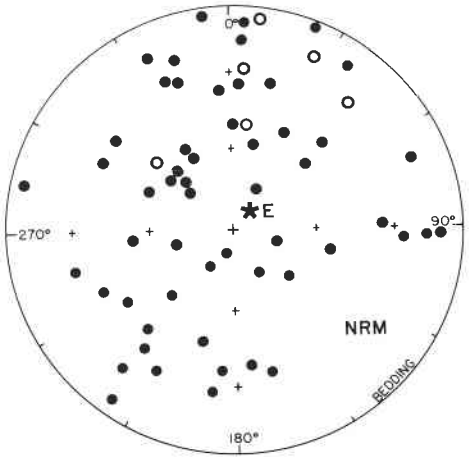


FIG. 2. NRM directions of specimens that yield the A_R direction. Solid (open) symbols depict directions with positive (negative) inclinations in this and succeeding stereographic projections (equal area). E is the direction of Earth's field in the general sampling area.

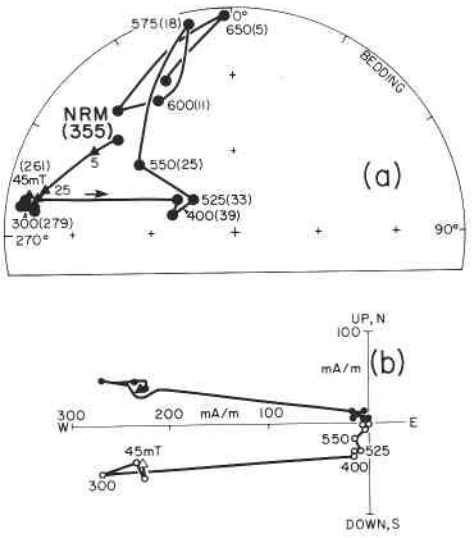


FIG. 3. Two-stage treatment of a specimen from site 4 illustrating the removal of an initial magnetization (C) by alternating fields (triangles), and the removal of at least three further magnetizations (A_N, A_R, and R_M) by thermal treatment to 660°C (circles). A_N is removed at some point below 300°C, A_R between 300 and 400°C, and R_M from 400 to about 575°C. (a) shows the directions, and (b), the vector diagram of the thermal demagnetization vectors. In (b) solid circles (or triangles) refer to the vector projection on the horizontal plane (N–E) and open circles (or triangles) to the vector projection on the east–west vertical plane (UP–E) (superimposed). Intensities in mA/m are shown in parentheses.

16°), they are not considered further. For convenience these magnetizations are labelled C.

The two-stage method of demagnetization generally consisting of low AF treatment between 20 and 45 mT followed by thermal demagnetization up to as high as 675°C was found to be the most effective procedure for uncovering the magnetizations present. Under low AF's the NRM direction typically migrated to a west direction of shallow inclination through the elimination of C (Figs. 3a, 4). The subsequent thermal demagnetization revealed this westerly magnetization to be composed of five magnetic components, which would have remained hidden without (or with only) the preliminary AF treatment (Fig. 5). Below 200–300°C VD's of several specimens reveal the elimination of a shallowly inclined easterly component (A_N) (Fig. 3b). The dominant component of the westerly magnetization (A_R) was removed in temperatures from about 200–450°C (Figs. 3, 4, 7). The resultant magnetization (R_M) after the removal of A_R had steeply inclined westerly (Fig. 3) to predominantly intermediately (Fig. 4) or shallowly inclined easterly directions. In contrast, high preliminary AF treatment tended to completely eliminate A_R (Fig. 7) with the subsequent thermal treatment occasionally revealing R_M to be in a shallowly inclined *westerly* direction (Fig. 8). For the low AF treated specimens at higher temperatures, between 500 and 575°C, the direction often migrated to a direction B_R similar to A_R (e.g., approximated by the 600°C point, Fig. 4). At still higher temperatures, above 575°C, B_R was occasionally eliminated to reveal a shallowly inclined easterly magnetization (B_N) similar in direction to A_N (not figured). Above 600°C, specimens often acquired a spurious magnetization. In summary, thermal demagnetization following low AF treatment indicates the presence of at least five magnetizations: two of them with similar westerly directions, A_R and B_R; two with similar easterly directions, A_N and B_N; and one, R_M, with a direction that falls roughly on a great circle passing through the two groups but predominantly lies in an intermediately inclined easterly direction (Fig. 9). Most of these components appear to be present in each specimen, though all of them are not resolvable.

Because its direction is rather scattered, the nature of R_M is unknown. The intensity decay curves often show a slight increase in intensity in the 400–500°C range following the elimination of A_R (Fig. 7), and the VD's indicate somewhat erratic behaviour in the extended range 400–550°C (Figs. 3, 4, 8). Even site 1, which yielded a rather coherent steep direction (Fig. 9), displays erratic behaviour in this temperature range (Fig. 10). These characteristics, together with the scattered direction, suggest that R_M is a mixed magnetization. The possible nature of R_M is discussed in the next section.

Judging by the magnetic characteristics found under treatment, three of the six components uncovered can be

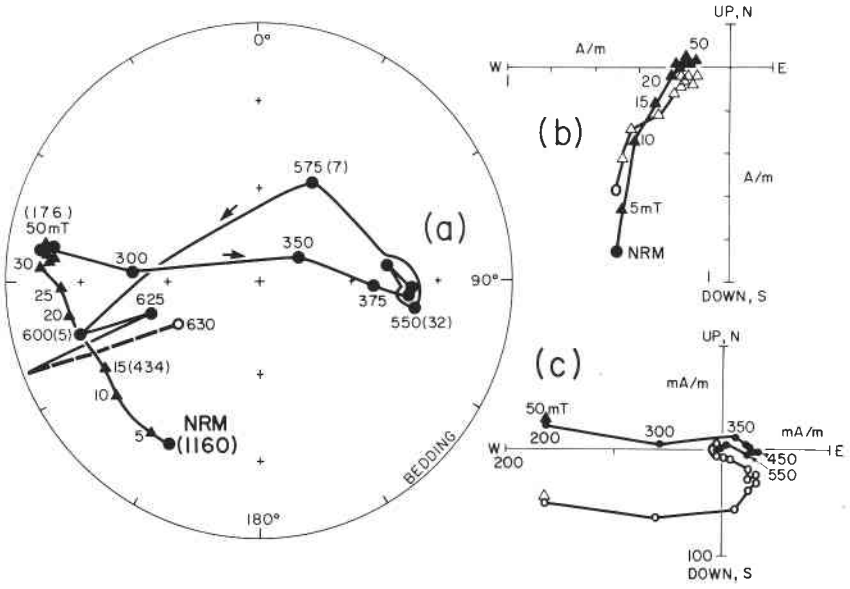


FIG. 4. Similar to Fig. 3, but this specimen from site 13 indicates the presence of an additional magnetization B_R with T_{UB}'s >575°C. B_R can be recovered by VS techniques, but in this example there are insufficient data available for its resolution. (*a*) depicts the directions, and (*b*) and (*c*), the vector diagrams of the AF and thermal data, respectively. See Fig. 3 for other explanations.

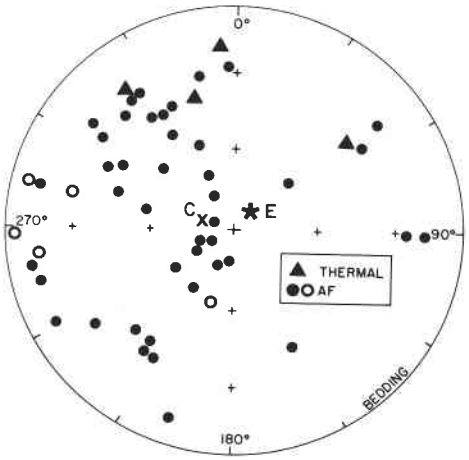


FIG. 5. *C* directions calculated from vector diagrams (Zijderveld 1967) of specimens treated by either heat or AF's but not both. C and E are the Cretaceous and present field directions in the general sampling area.

related in a general way to the observed presence of magnetite and hematite in the polished thin sections (one from each site). The relatively high intensities associated with T_{UB}'s around 575°C point to magnetite as the carrier of the R_M magnetization, whereas the low intensities associated with T_{UB}'s above 575°C indicate hematite as the carrier of B_R and B_N. Specimens from site 1 show a sharp drop in intensity at about the Curie point of magnetite (Fig. 10).

Magnetite occurs in at least three forms: as relatively fresh grains, as grains with a cruciform growth pattern indicative of quenching conditions, and as large altered grains up to 3000 μm with trellis, sandwich, and composite intergrowths of unaltered ilmenite. It may additionally occur as very fine particles in the highly altered silicate matrix, but the occasionally numerous small opaques could not be identified. The predominant intergrown form contains minor hematite alterations along fractures as well as major maghemite alterations affecting as much as 30% of grain interiors. These latter alterations, although not microscopically identified, present a pattern of fine granulation associated with irregular cracks, which is typical of low-temperature oxidation of magnetite to produce maghemite, where the cracks are evidence for the more compact maghemite structure. The abundance of ilmenite in these grains implies that the magnetite is either pure or of rather low titanium content, and hence that its oxidation product is maghemite rather than titanomaghemite. The physical evidence for maghemite in these large grains would suggest that some of the small opaques in the matrix are also maghemite grains produced under similar conditions.

The remaining two components uncovered (A_R, A_N) probably reside in the maghemite. Maghemite inverts to hematite between 250 and 550°C (Haggerty 1976, p. Hg-135). The magnetic characteristics of A_R with T_{UB}'s from 200 to 450°C or so and high intensities in addition to the physical evidence are especially indicative of

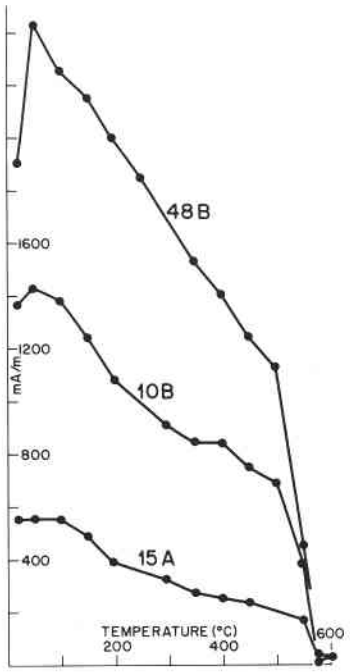


FIG. 6. Typical thermal decay curves of specimens (from sites 39(48B), 2(10B), and 4(15A)) having no prior AF treatment.

maghemite (Figs. 3, 4, 7). The magnetic characteristics of A_N are less certain, but since its direction is antiparallel to A_R within error (Table 3) and its T_{UB} range is largely different from that of A_R, perhaps owing to a reversal of the Earth's magnetic field during the cooling of the sill, it is suggested that it is also carried by maghemite. It could alternatively be carried by goethite or pyrrhotite. Goethite was not identified in the polished sections, but is an unlikely possibility because any goethite formed earlier would almost certainly have been dehydrated during the subsequent deep burial between early Cambrian and Cretaceous times. However, pyrrhotite was tentatively identified, and any magnetization harboured by single domain (sd) grains would have survived the low AF treatment to 45 mT (Soffel 1977).

The magnetizations have been resolved by the use of VD's or VS techniques. B_R and B_N have been calculated exclusively by VS and A_N exclusively by the use of VD's, but both methods have been used for resolving A_R. There is excellent agreement between the mean A_R direction calculated by each method separately from the same specimens (compare entries 5 and 6, Table 4). In calculating final results, directions derived by both methods have been averaged together at the specimen level. R_M has been roughly determined by subtracting from the approximate end-point vector the small magnetization remaining after the elimination of R_M. All individual specimen directions except those of A_N are

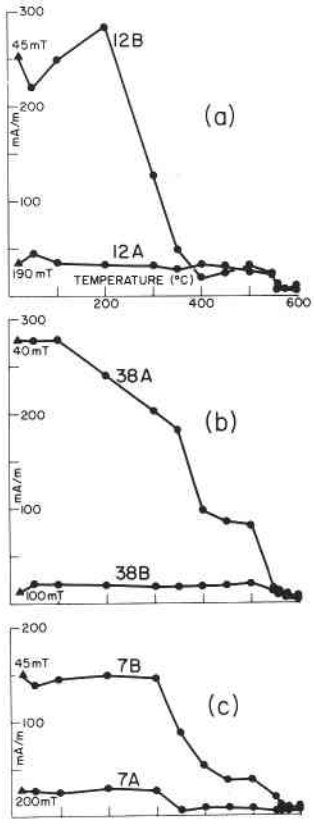


FIG. 7. Examples of thermal demagnetization of specimens from the same core comparing the effect of different preliminary AF treatments. Companion specimens from sites 3(*a*), 37(*b*), and 2(*c*) are shown. Note the slight increase in some of the curves following the elimination of A_R at about 400°C, which indicates the removal of a possible antiparallel magnetization. B_R and B_N occupy the T_{UB} range beyond about 550°C.

plotted after bedding correction in Fig. 9. Tables 2 and 3 give the A_R and B directions by site with specimen unit weight, and Table 4 is a summary of all directions calculated in a variety of ways. In this study the author prefers the final averages that have been calculated using specimen rather than site unit weight because of the wide variation in the number of resolved specimen directions per site, but the average calculated from site unit weight after the removal of one-specimen sites is also quoted (entries 1 and 2). There is little difference in the direction calculated by either method. There is little difference too between the *in situ* and the bedding corrected directions; however, the latter are preferred since the directions appear to be prefolding (see below).

Discussion and summary

In this study two-stage treatments were essential in uncovering the A_R, A_N, and R_M magnetizations. Low AF's in the first stage generally tended to enhance the

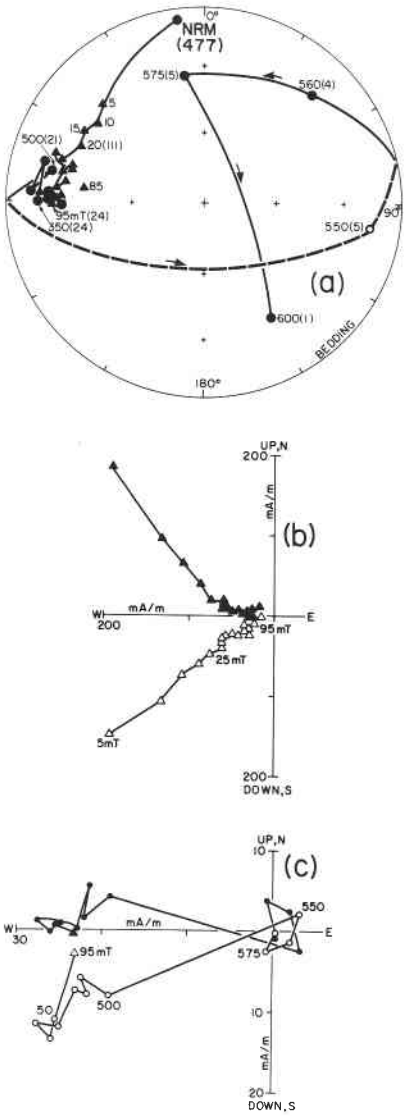


FIG. 8. Illustration of thermal demagnetization of a specimen from site 4 following treatment in relatively high alternating fields. A_R has been largely eliminated in 95 mT, leaving predominantly R_M in a shallow westerly direction. (*a*) depicts the directions, and (*b*) and (*c*), the vector diagrams of the AF and thermal data, respectively. See Figs. 2 and 3 for other explanations.

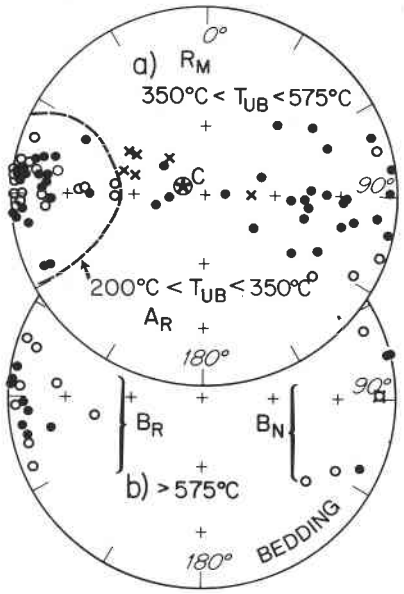


FIG. 9. A_R and R_M (*a*) and B_R and B_N (*b*) directions from thermal demagnetization of individual specimens previously treated by alternating fields. The directions have been resolved using vector subtraction and vector diagrams. Crosses depict site 1 directions and the symbol denoted by C shows the Cretaceous field direction in the sampling area.

presumed maghemite component (A_R) relative to the magnetite (R_M) as indicated by the large intensity decay in the 200–400°C range after subsequent thermal treatment (compare Fig. 6 with Fig. 7), whereas high AF's in the first stage generally had the opposite effect, completely removing the maghemite component (Fig. 7*a*, *b*). Sometimes both components were removed in nearly equal proportions with both low and high AF's (Fig. 7*c*), thus indicating the variable ranges of the coercivity spectra of these minerals. Site 1 was unique after low AF two-stage treatment in displaying thermal intensity decay curves that were almost solely representative of magnetite rather than maghemite (Fig. 10). Though A_R is usually enhanced relative to R_M under low AF, its determination will be slightly affected by the simultaneous removal of R_M. The biasing effect of R_M, however, should be negligible, not only due to the usually low magnetite/maghemite intensity ratios but to the different ranges of the main T_{UB}'s of each component (Fig. 7).

The exact nature of the magnetite magnetization R_M is unknown. Judging by the vector diagrams of the thermal results from two-stage treatment (including those for site 1; Fig. 10), R_M is a mixed component or vector resultant with individual specimen directions ranging from the vicinity of A_N to the vicinity of A_R. This evidence thus suggests that R_M is composed of normal and reverse phases having the A_R/B_R and A_N/B_N directions. The actual distribution of directions hints that it may include an additional component having the Cretaceous direction. This is especially suggested by the coherent direction near the Cretaceous of the most extensively altered site (1) and by the mean R_M direction, which together with A_R falls on a great circle passing within several degrees of the Cretaceous direction (Fig. 9). This interpretation, though, rests partly on the presence of site 1, which comes from a different and possibly non-contemporaneous sill. A Cretaceous component

TABLE 2. A_R site vectors after bedding correction

Si	Coordinates Long. (°W) Lat. (°N)	Sp	Int	D(°), I(°)	R	k	α_{95}(°)	Pole Long.(°) Lat.(°)
1	127.99 64.163	—	—	—	—	—	—	— —
2	127.96 64.152	7	59	271, +06	6.753	24	13	143E 03N
3	127.96 64.152	1	299	(275, −01)	—	—	—	(137E 02N)
4	127.96 64.152	6	154	278, +12	5.941	85	7	138E 08N
5	127.96 64.152	4	178	272, −05	3.937	48	13	139E 02S
13	127.96 64.157	6	164	272, +03	5.857	35	12	141E 02N
22	128.15 65.02	—	—	—	—	—	—	— —
24	128.14 65.03	4	51	282, −02	3.958	72	11	131E 04N
37	131.01 65.20	1	39	(270, −16)	—	—	—	(135E 07S)
38	131.03 65.24	3	117	(277, −08)	2.444	—	—	(131E 00N)
39	131.34 65.22	1	38	(273, −37)	—	—	—	(127E 17S)
40	131.34 65.27	5	50	256, −08	4.646	11	24	150E 10S

NOTES: Si is the site number; Sp is the number of specimens; Int is the intensity (mA/m); D°, I° are the declination and inclination of the direction; R is the vector resultant; k is the precision; and α_{95}° is the error of 95% confidence about the direction. Values in parentheses are not used in calculating the site average quoted in Table 4 (entry 2). The maximum errors in longitude and latitude for sites 1–5 and 13 are 110 and 56 m, respectively, for sites 22 and 24, 250 and 250 m, and for sites 37–40, 375 and 900 m.

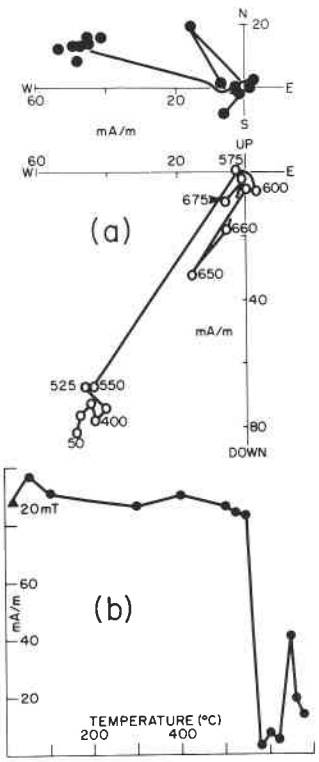


FIG. 10. Vector diagram (*a*) and intensity decay curve (*b*) of the thermal results of one specimen from site 1 previously treated in alternating fields. See Fig. 9 for cleaned directions of site 1 and Fig. 3 for explanation of the vector diagram.

could have been caused by the Laramide uplift and folding in the area. In the nearby Little Dal Group this orogeny appears to have produced a magnetization carried by goethite (Park 1981), but in the sills the genesis of the magnetite magnetization is problematical.

A_N and A_R are essentially single components that are antiparallel within error (Table 4) and have largely non-overlapping T_{UB} ranges. From this limited evidence A_N, like A_R, would appear to be carried by maghemite. Another possibility is that it is carried by sd pyrrhotite. A_R is the major component recovered by this study, and through arguments (1) and (2) below it is suggested to be a *primary* magnetization formed by *deuteric* alteration.

(1) In nature, maghemite is usually found as an alteration product of magnetite. It is also possible for it to form during the initial stages of an intrusion by quite limited solid solution of magnetite and hematite above 1000°C (Lindsley 1976). However, from the physical evidence in this study it is suggested that the maghemite formed at much lower temperatures from magnetite either by deuteric alteration or by later weathering. Either of these latter two processes acting on sd or pseudo-single-domain (psd) grains of magnetite should produce maghemite, which retains the internal field direction of the magnetite owing to positive exchange coupling or shape anisotropy (Johnson and Merrill 1974). On the other hand, any multi-domain (md) grains of maghemite that are formed will record the external field (Johnson and Merrill 1974), though any magnetization caused by this source in the present study should have been eliminated by the AF treatment. The first

TABLE 3. *B* site vectors after bedding correction

Si	Sp	Int	$D(°)$, $I(°)$	R	k	$\alpha_{95}(°)$	Pole Long.(°) Lat.(°)
1	1	3	269, +10	—	—	—	145E 04N
2	1	9	268, +09	—	—	—	146E 03N
3	1	6	267, −05	—	—	—	144E 04S
4	—	—	—	—	—	—	— —
5	1	8	260, −43	—	—	—	139E 27S
13	3	19	264, +06	2.88	—	—	148E 00N
22	2	32	280, +12	1.96	—	—	135E 09N
24	3	33	260, +04	2.93	—	—	152E 02S
37	—	—	—	—	—	—	— —
38	1	12	264, −04	—	—	—	143E 04S
39	4	19	273, −09	3.84	18	22	134E 03S
40	1	10	291, −08	—	—	—	118E 05N

NOTES: See Table 2 for explanation of symbols. These directions have been calculated using only vector subtraction techniques (Park and Roy 1979). Where applicable normal and reverse directions have been combined within specimens and quoted as reverse directions.

TABLE 4. Summary of vector directions and poles

	Unit weight, description	N	Int	Directions $D(°)$, $I(°)$	R	k	$\alpha_{95}(°)$	Poles Long.(°) Lat.(°)	R	K	$A_{95}(°)$	$\delta p(°)$, $\delta m(°)$
1. A_R	Si, *in situ*	6	109	273, −01	5.88	40	11	139E 01N	5.93	75	8	5, 11
2.	Si, bedding	6	109	272, +01	5.90	50	10	140E 01N	5.94	85	7	5, 10
3.	Sp, *in situ*	38	111	273, −01	35.72	16	6	— —	—	—	—	3, 6
4.	Sp, bedding	**38**	**111**	**272, +00**	**35.76**	**17**	**6**	**139E 01N**	**36.92**	**34**	**4**	**3, 6**
5.	Sp, VD	33	—	271, +00	30.69	14	7	— —	—	—	—	4, 7
6.	Sp, VS	33	—	271, −02	30.95	16	7	— —	—	—	—	3, 7
7.	Sp, ≤ 45 mT	25	117	272, +04	24.10	27	6	— —	—	—	—	—
8.	Sp, >45 mT	13	98	273, −09	11.86	11	13	— —	—	—	—	—
9. A_N	Sp	7	36	089, −08	6.73	22	13	— —	—	—	—	—
10. R_M	Sp, all	30	52	086, +55	21.33	3	17	— —	—	—	—	—
11.	Sp, site 1	6	93	300, +66	5.66	15	18	144E 55N	—	—	—	—
12. B_R	Sp	17	20	270, −03	16.06	17	9	141E 02S	16.35	25	7	4, 9
13. B_N	Sp	6	9	090, −07	5.41	9	24	141E 04N	5.50	10	22	12, 25
14. *B*	Sp	**18**	—	**269, −01**	**17.02**	**17**	**9**	**141E 01S**	**17.38**	**27**	**7**	**4, 9**
15.	Directions	23	—	270, −01	21.40	14	9	141E 01S	21.83	19	7	4, 9

NOTES: Data have been derived from the thermal demagnetization of specimens previously treated in alternating fields. All directions are corrected for bedding unless otherwise indicated. Data in boldface are considered to be the best summaries of the data. N is the number of sites (Si) or specimens (Sp); Int is the intensity (mA/m); $D°$, $I°$ are the declination and inclination; R is the vector resultant; k or K is the precision; $\alpha_{95}°$ or $A_{95}°$ is the error of 95% confidence about the direction or pole; and $\delta p°$, $\delta m°$ are the semi-axes of the ellipse of confidence at $P = 0.05$ about the pole calculated from the average directions. Where individual specimen directions have been calculated using both VD and VS, the values have been averaged. They are also quoted separately here for comparison (entries 5 and 6). Also quoted are the mean directions of those specimens previously treated in AF's ≤ 45 mT (entry 7) and those in AF's >45 mT (entry 8) prior to thermal demagnetization.

conclusion then is that A_R should indicate the original magnetite direction whether or not A_R results from deuteric alteration or weathering.

(2) The alteration process on sd or md grains should show no preference for either normally or reversely magnetized grains; therefore these grains should be altered in the same proportion as their occurrence. It is apparent that R_M is predominantly biased in the normal direction, 125° from A_R. Hence, the second conclusion is that A_R was formed deuterically prior to the presumed normal magnetite component. Conclusions (1) and (2) therefore imply that A_R is a primary magnetization.

The primary nature of A_R is supported by directional agreement with the presumed hematite B_R and B_N magnetizations, which are antiparallel to each other within error and within 2° of A_R (Table 4). However, one cannot eliminate the possibility that the hematite originated by later weathering rather than by deuteric alteration, and that is was magnetized in the fields of *A* and *R*. In fact some of the hematite could have originated during the thermal experiments through the inversion of maghemite; however, the intensity of the observed components is far too high for the bulk of it to be accounted for by this effect.

The poles of *A* and *B* are given in Table 4. On first consideration it would seem likely that these poles are *virtual* geomagnetic poles (vgp) or spot readings of the direction of the Earth's magnetic field, because all of the useful results were obtained from what appears to be a single sill that presumably cooled within a relatively short time compared with the secular variation period of the Earth's field. Calculations made using the method and basic assumptions of Thorning and Abrahamsen (1980) appear to confirm this contention, for they indicate that the 20–60 m thick sill of this study buried at a depth of at least 3.5 km would have cooled through the main A_R T_{UB} interval of 400–300°C in a period of 100–200 years as compared with a probable duration of the secular variation cycle of roughly 5000–20 000 years. There are several arguments against this idea of vgp(s). First, the sampling localities extend over a distance of some 200 km, and it is difficult to imagine that the events at each locality were contemporaneous, especially when considering such probable dynamic processes as multiple intrusions. Second, there are three or more sills present in some areas, which indicates that intrusion of magma took place over an extensive period. Third, the dual polarity of both *A* and *B* indicates that the remanence acquisition process spanned field reversal time scales (10 000 years). It is therefore suggested that any bias owing to secular variation has been somewhat mitigated.

Readers are referred to Morris and Park (1981) for a discussion of these poles in relation to other poles of the Mackenzie Mountains and the North American craton.

Acknowledgements

I would like to express my appreciation to J. D. Aitken of the Geological Survey of Canada for his original suggestion of this project, for his provision of camp facilities and logistical support, and for his valuable geological advice. Mr. S. Savard provided valuable field assistance. Criticism of the manuscript by P. Lapointe and J. L. Roy of the Paleomagnetic Section of the Earth Physics Branch and by W. A. Morris of Morris Magnetics was much appreciated. The manuscript was further improved as a result of editing suggestions from the journal reviewers. I am also grateful to Mrs. Jean Hastie for help in specimen measurement and data handling.

AITKEN, J. D. 1980. Sedimentation of the Helikian Little Dal Group, Mackenzie Mountains. Proterozoic Basins in Canada Symposium, Geological Association of Canada meeting, Halifax, N.S., Program with Abstracts.

AITKEN, J. D., and COOK, D. G. 1974. Carcajou Canyon map area, District of Mackenzie, Northwest Territories. Geological Survey of Canada, Paper 74-13, pp. 1–28.

HAGGERTY, S. E. 1976. Opaque mineral oxides in terrestrial igneous rocks. *In* Oxide minerals. *Edited by* D. Rumble, III. Mineralogical Society of America, Vol. 3, pp. Hg-101–Hg-300.

IRVING, E., PARK, J. K., and EMSLIE, R. F. 1974. Paleomagnetism of the Morin complex. Journal of Geophysical Research, **79**, pp. 5482–5490.

JOHNSON, H. P., and MERRILL, R. T. 1974. Low-temperature oxidation of a single-domain magnetite. Journal of Geophysical Research, **79**, pp. 5533–5534.

LINDSLEY, D. H. 1976. The crystal chemistry and structure of oxide minerals as exemplified by the Fe–Ti oxides. *In* Oxide minerals. *Edited by* D. Rumble, III. Mineralogical Society of America, Short Course Notes, Vol. 3, pp. L-1–L-25.

MORRIS, W. A., and PARK, J. K. 1981. Correlation of Upper Proterozoic strata in the Cordillera: paleomagnetism of the Tsezotene sills and the Little Dal lavas. Proterozoic Basins in Canada Symposium, Geological Association of Canada meeting, Halifax, N.S.

PARK, J. K. 1981. Analysis of the multicomponent magnetization of the Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada. Journal of Geophysical Research, **86**. (In press.)

PARK, J. K., and ROY, J. L. 1979. Further paleomagnetic results from the Seal Lake Group igneous rocks, Labrador. Canadian Journal of Earth Sciences, **16**, pp. 895–912.

ROY, J. L., ROBERTSON, W. A., and KEEPING, C. 1969. Magnetic "field free" spaces for paleomagnetism, rock magnetism and other studies. Canadian Journal of Earth Sciences, **6**, pp. 1312–1316.

ROY, J. L., REYNOLDS, J., and SANDERS, E. 1973. An alternating field demagnetizer for rock magnetism studies. Publications of the Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa, Ont., No. 44(3), pp. 37–45.

SOFFEL, H. 1977. Pseudo-single-domain effects and single-domain–multidomain transition in natural pyrrhotite deduced from domain structure observations. Journal of Geophysics, **42**, pp. 351–359.

THORNING, L., and ABRAHAMSEN, N. 1980. Paleomagnetism of Permian multiple intrusion dykes in Bohuslan, SW Sweden. Geophysical Journal of the Royal Astronomical Society, **60**, pp. 163–185.

ZIJDERVELD, J. D. A. 1967. A.C. demagnetization of rocks: analysis of results. *In* Methods in paleomagnetism. *Edited by* D. W. Collinson, K. M. Creer, and S. K. Runcorn. Elsevier Publishing Co., New York, NY, pp. 254–286.